

TrES-5b

R-0906

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250701 · created 2026-07-29T05:20:05+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460857.8924 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.012
Transit depth (Rp/Rs)^2	3.4 +/- 0.42 [%]
Orbital Inclination (inc)	84.83 +/- 1.81
Airmass coefficient 1 (a1)	1.35 +/- 0.03
Airmass coefficient 2 (a2)	-0.26 +/- 0.13
Scatter in the residuals of the lightcurve fit is	2.25 %
Best Comparison Star	#2 - [270, 279]
Optimal Method	PSF photometry
Transit Duration (day)	0.0785 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.184 ± 0.012 vs 0.142 (deviation 3.5σ)
Duration fit vs published	0.0785 d vs 0.0758 d (deviation 0.29σ)
Mid-time O–C	-11.0 min
Depth SNR	15.3
Residual scatter	2.25 %

Light curve

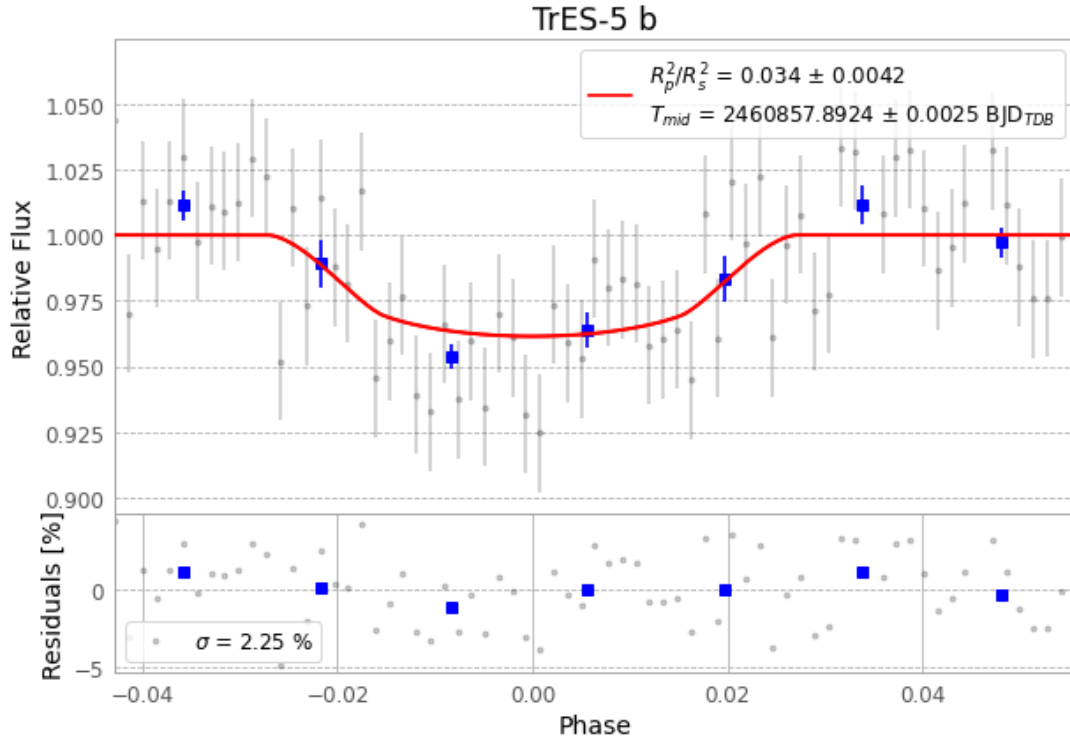


Figure 1 — FinalLightCurve_TrES-5 b_2025-07-01.png (SHA-256 53739e00d84d5e3e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 245], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c2d40efc0f65a0bf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

71 FITS frames (47 MB), TRES-5250701075115.FITS → TRES-5250701112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250701.log (df4b0a776586...), FinalLightCurve_TrES-5 b_2025-07-01.png (53739e00d84d...), FinalLightCurve_TrES-5 b_2025-07-01.csv (c330f409c438...)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 05:20Z.